Energy-Efficient Llama3 Acceleration on a CGLA by Offloading Computational Kernels

Takuto ANDO, Yu ETO, Ayumu TAKEUCHI and Yasuhiko NAKASHIMA
Nara Institute of Science and Technology, 8916-5 Takayama-cho, Ikoma, Nara 630-0192, Japan.

***Abstract*—The evolution of Large Language Models (LLMs) has led to a substantial increase in energy consumption, making energy-efficient accelerator technologies essential. We propose a method to maximize the performance of the Llama3 LLM on IMAX3, a CGRA-based accelerator. We conducted a detailed analysis of the Q3_K_S quantization method. Based on this analysis, we implemented a new custom kernel to offload the Q6_K operations to the hardware. In previous work, this operation was processed on the host CPU. This targeted optimization reduced the end-to-end latency by approximately 11%. Furthermore, an evaluation of a projected ASIC implementation showed that the absolute performance does not match that of a GPU. In contrast, our approach achieves superior power efficiency, as measured by the Power-Delay Product (PDP).**

***Index Terms*—Llama, LLM, CGRA, CGLA, IMAX**

I. Introduction

Recent advancements in AI, including Large Language Models (LLMs), rely on Graphics Processing Units (GPUs), leading to a major increase in energy consumption and necessitating new, power-efficient accelerators. Akabe et al. have proposed IMAX, a Coarse-Grained Reconfigurable Array (CGRA)-based accelerator architecture designed for high power efficiency and flexibility [1]. Furthermore, Eto et al. executed a Llama2-based model on IMAX3 and showed that it could achieve higher power efficiency compared to a GPU under specific conditions [2]. However, this evaluation had two main limitations. First, the model used was from a previous generation. Second, during the assessment of the Q3_K_S quantized model, only the primary 3 bit operations (Q3_K) were offloaded to the CGRA.

Through our analysis of the Q3_K_S quantization implementation, we found that 6 bit operations (Q6_K) are executed alongside the main 3 bit operations. These operations make up approximately 10 % of the total workload. In previous work, these 6 bit operations were processed on the host CPU, offering an opportunity to improve both the offload ratio and overall performance. Therefore, this paper introduces two contributions: First, we update the evaluation model to the latest Llama3 to reassess the performance of IMAX3. Second, we propose and implement a new custom kernel (Q6_K) to offload the 6 bit operations, which were previously executed on the host CPU, directly to the IMAX.

II. Proposed method

A. IMAX3 CGRA Architecture

IMAX3 is a multi-lane CGRA accelerator designed for high data locality and power efficiency [1]. Its architecture consists of a linear array of Processing Elements (PEs) and on-chip Local Memory Modules (LMMs). Each lane operates in a dataflow fashion, where data is streamed from LMMs through a chain of PEs, minimizing long-distance data movement and the overhead of a complex instruction pipeline.

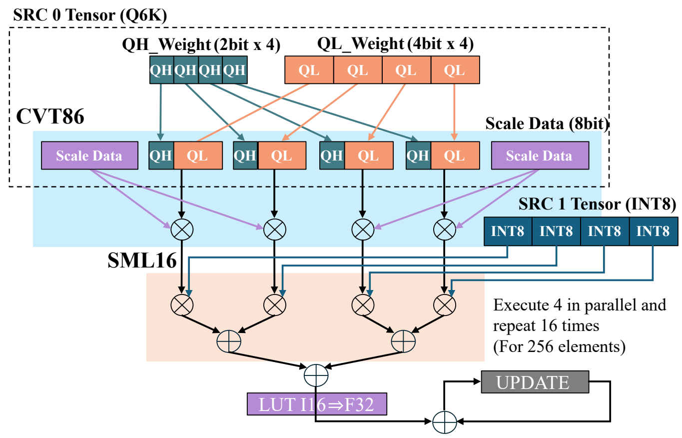

Fig. 1: Dataflow of the Q6_K kernel

B. Offloading the Q6_K Kernel to the CGLA

The Q3_K_S quantized method, as used in Llama.cpp, is a technique that reduces model size to an equivalent of 3 bit per weight. In previous work, only the primary 3 bit dot-product operations (Q3_K) were offloaded to the IMAX3. However, our detailed analysis of the Q3_K_S model implementation revealed that 6 bit dot-product operations (Q6_K), which account for approximately 10 % of the total operator calls, were still being executed on the host CPU. Therefore, we have designed and implemented a new kernel to execute these Q6_K operations directly on IMAX. Fig. 1 illustrates a portion of the dataflow for the proposed kernel. Within the Q3_K_S format, the weights for Q6_K (the src0 tensor) are structured in blocks composed of 2 bit and 4 bit quantized weights, along with their corresponding 8 bit scale data. The other input (the src1 tensor) is an 8 bit integer (INT8).

The processing within the proposed kernel consists of two stages. First, a custom instruction named CVT86 decodes the 2 bit and 4 bit weight data from src0 using the 8 bit scale data to generate 16 bit intermediate data. Second, another custom instruction, SML16, performs a dot-product operation between this decoded 16 bit intermediate data and the 8 bit integer input from src1. This custom kernel enables efficient

TABLE I: Physical specifications and performance comparisons of various devices

Device	CPU	Cores	Chip area (mm^2)	Process node (nm)	Operating frequency (MHz)	Memory	Power (W)
IMAX3 (Xilinx VPK180)	ARM Cortex-A72	64[a]	-	7	140	8 GB + 4 GB DDR4	180
IMAX3 (28nm)	-	64[a]	14.6	28	840	-	25.4 or 31.7[c]
Intel i9-12900KF	-	16	-	7	2400	512 GB DDR5	125
NVIDIA GTX 1080 ti	i9-12900K	3584[b]	471	16	1480	11 GB GDDR5X	250

[a] The number of cores for IMAX3 refers to the number of PEs per lane.
[b] The number of cores for the NVIDIA GTX 1080 ti refers to the number of CUDA cores.
[c] The power consumption for the IMAX3 (28 nm) is an estimated value for the configuration with 256 KB LMM.

on-hardware processing of Q6_K operations, eliminating the need for host CPU intervention. The IMAX architecture provides two parallelization mechanisms, SIMD and column-wise multithreading, both of which are fully utilized in our kernel implementation. For example, to compute a 256 elements dot product, the kernel achieves a high degree of parallelism by executing four 4 elements operations concurrently using SIMD and repeating this process 16 times, as depicted in Fig. 1. This optimization increases the hardware offload ratio from approximately 90 % in the previous approach to maximizing the utilization of the IMAX3's parallel processing capabilities. We aim to improve the overall system performance.

III. Experimental Evaluation

We evaluated the end-to-end performance and power efficiency of our method to compare the IMAX executing the latest Llama3 model against general-purpose devices, and to quantify the improvement from offloading the Q6_K kernel. As detailed in Table I, our evaluation platform includes three types of device. For the IMAX3, we used both an FPGA prototype operating at 140 MHz and a projected performance figure for an ASIC implementation on a 28 nm process, estimated to run at 840 MHz. We configured the local memory to 256 KB, as the maximum 512 KB was larger than required for this workload. For the GPU comparison, we selected the NVIDIA GTX 1080 ti. We chose this model to enable a fair architectural comparison, as its 16 nm process node is comparable to the 28 nm node of projected IMAX3 (ASIC). The workload was the Q3_K_S quantized version of the Llama3-8B model. We evaluated end-to-end latency for generating 32 tokens and the Power-Delay Product (PDP), an energy efficiency metric defined as $\mathrm{PDP} = \mathrm{Latency} \times \mathrm{Power}$.

First, we compared the IMAX3 (ASIC) with the implemented Q6_K kernel against the GPU and CPU. As shown in Fig 2, the results indicate that the GPU has superior absolute performance with both one and two threads. However, as illustrated in Fig 3, IMAX3 substantially outperforms the GPU in terms of power efficiency. This demonstrates that the IMAX3 architecture maintains a considerable advantage in energy efficiency, even when running the latest Llama3 model.

Next, we verified the improvement achieved by offloading the Q6_K kernel. By offloading this kernel, the end-to-end latency was reduced from 34.2 s to 30.4 s, an improvement of approximately 11 %. The offloaded Q6_K operations constitute only about 10 % of the total operator calls in the entire computation. This performance improvement of over 10 %

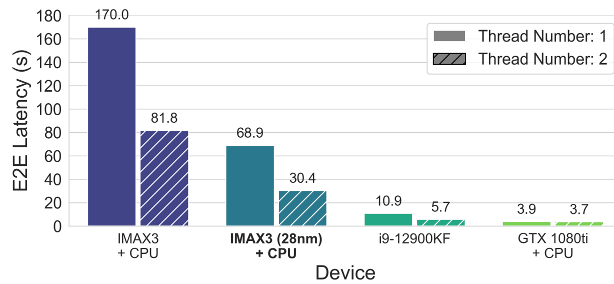

Fig. 2: E2E latency comparison with other devices

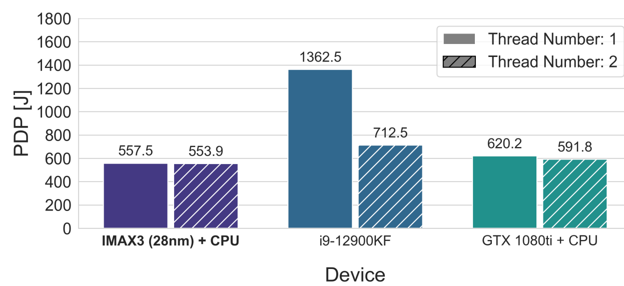

Fig. 3: PDP comparison with other devices

highlights the inefficiency of the previous host-CPU execution and underscores the effectiveness of our targeted optimization.

IV. Conclusion

In this paper, we evaluated and optimized the performance of the IMAX3 CGRA accelerator for Llama3. We implemented a new custom kernel to offload the 6 bit operations (Q6_K) that are present within the Q3_K_S quantized model. Our evaluation demonstrated that this optimization increased the hardware offload ratio and improved end-to-end performance by approximately 11 %. Furthermore, a comparison with GPU, CPU, and IMAX3 (ASIC) showed that our IMAX3 achieves superior energy efficiency. For future work, we will investigate application-specific architectural parameters, such as the number of processing elements and the LMM capacity.

References

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